ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	The properties of the convex mirror explain its use.
2	**(c)** 11.25 cm	1/f = -1/45 - 1/15 = -4/45, so f = -11.25 cm.
3	**(a)** convex, f = 12 cm	Object at 2F.
4	**(a)** Both A and R are true and R is the correct explanation of A.	Refracted rays only appear to meet when extended backwards.
5	**(a)** -40 cm	m = -v/u = 2 gives v = +40; 1/f = 1/40 - 1/20 = -1/40.
6	**(c)** 1.5 f	1/v = 1/f - 1/3f = 2/3f.
7	**(d)** A is false but R is true.	A is false (image is virtual); R is true.
8	**(b)** decreases	Magnification approaches 1.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The wide field of view of the diminished, erect image lets the driver see more traffic behind.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the ray falls normally on the mirror and retraces its path, which is not explained by $R = 2f$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	m = -1/3 = -v/u, v = -15, so u = -45 cm (1); 1/f = -1/15 - 1/45 = -4/45, f = -11.25 cm (1).	1+1
12	Image of the Sun is at the focus, f = 20 cm (1); R = 2f = 40 cm (1).	1+1
13	Ray hits perpendicular to the surface (1); No bending and no sideways displacement (1).	1+1
14	Lens: 1/v = 1/20 - 1/40, v = +40 cm (1); Mirror: 1/v = -1/20 + 1/40, v = -40 cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	First case: u = -x, v = -3x, so 1/f = -1/(3x) - 1/x = -4/(3x), |f| = 3x/4 (1); Second case: same size means the object is at C, so x + 10 = 2|f| = 3x/2 (1); x = 20 cm, so f = -15 cm (1).	1+1+1
16	m = -5, v = -5u (1); 1/12 = 1/(5|u|) - ... with u negative: 1/12 = 1/(5|u|) + 1/|u| = 6/(5|u|) (1); |u| = 14.4 cm, v = 72 cm (1).	1+1+1
17	Water is rarer than glass, so the ray bends away from the normal (1); n(w,g) = 1.33/1.5 (1); = 0.887 (1).	1+1+1
	OR: m = v/u = -3, v = 80, u = -26.7 cm (1); 1/f = 1/80 + 1/26.7 (1); f = 20 cm (1).	1+1+1
18	1/v = 1/2 + 1/4 = 3/4, v = +1.33 m (1); m = -v/u = 1.33/4 (1); m = 0.33 (image is one-third) (1).	1+1+1
	OR: Angle of incidence is the angle with the normal = 0 (1); Angle of reflection = 0 (1); The ray retraces its path (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $f = R/2 = -20$ cm (1). (ii) $u = -30$, $f = -20$: $\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = -\frac{1}{20} + \frac{1}{30} = -\frac{1}{60}$, so $v = -60$ cm, i.e. 60 cm in front of the mirror (1). (iii) $m = -\frac{v}{u} = -\frac{-60}{-30} = -2$ (1). (iv) Image height $= 2 \times 3 = 6$ cm (1); real, inverted and enlarged (1).	1+1+1+2
20	1/u = 1/v - 1/f = 1/45 - 1/15 = -2/45 (1); u = -22.5 cm (1); m = v/u = -2; image 12 cm (1); Object at 45 cm: 1/v = 1/15 - 1/45 = 2/45 (1); v = 22.5 cm (1).	1+1+1+1+1
	OR: n_A = 3×10^8 / 2.4×10^8 = 1.25 (1); n_B = 3/1.5 = 2 (1); n(B,A) = 2/1.25 = 1.6 (1); B is denser (1); Ray bends towards the normal (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).

- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.
- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.
- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.
- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.
- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 2	n = 3/1.5.
2	**(c)** is complete but dimmer	Every part of the lens forms the entire image.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Refraction due to speed change.
4	**(d)** bends towards the normal	B is denser than A.
5	**(a)** has zero power and behaves like a plane glass plate	Net power is zero.
6	**(c)** A is true but R is false.	A is true; R is false: light is fastest in vacuum.
7	**(d)** 1.25	1.5/1.2.
8	**(a)** 12 cm	P = 5 + 3.33 = 8.33 D; f = 12 cm.
9	**(c)** A is true but R is false.	Glass is optically denser: light slows down and bends towards the normal; R is false.
10	**(d)** A is false but R is true.	A is false: a convex lens forms a virtual image for an object between O and F; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Virtual rays do not actually meet (1); Only real image has actual intersection of rays (1).	1+1
12	Let u = -x and v = +3x (1); 1/20 = 1/(3x) + 1/x = 4/(3x), so x = 26.7 cm (1).	1+1
13	Refractive index is more for violet than for red (1); Violet bends more, giving smaller f (1).	1+1
14	1/v = 1/10 - 1/20 = 1/20, v = 20 cm (1); Real, inverted, same size (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Frequency is fixed by the source (1); Speed depends on the medium (1); Wavelength changes as v = f x wavelength (1).	1+1+1
16	P = 2 - 0.5 = 1.5 D (1); f = 1/1.5 = 0.667 m = 66.7 cm (1); Convex (converging) (1).	1+1+1
17	f = -15 cm; 1/v = -1/15 + 1/20 = -1/60 (1); v = -60 cm, m = -3 (1); Image size = 15 cm inverted (1).	1+1+1
	OR: Apparent thickness = 6/1.5 = 4 cm (1); Shift = 6 - 4 (1); 2 cm (1).	1+1+1
18	Concave mirror image depends on position relative to F (1); Convex mirror rays always diverge (1); So only virtual images form (1).	1+1+1
	OR: A convex lens gives real image but suffers colour fringes (1); Combination reduces defects (1); Combination also improves image sharpness (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $f = \frac{1}{P} = \frac{1}{2.5} = 0.4$ m $= 40$ cm (1). (ii) $\frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{1}{40} - \frac{1}{80} = \frac{1}{80}$, so $v = +80$ cm (1). (iii) $m = \frac{v}{u} = \frac{80}{-80} = -1$ (1). (iv) Real, inverted and of the same size (1); for a magnified real image she should place it between F and $2F$, i.e. between 40 cm and 80 cm (1).	1+1+1+2
20	(a) 1/v = 1/f - 1/f = 0, so v is at infinity (1); (b) u = f/2 (both negative), 1/v = 1/f - 2/f = -1/f in magnitude form, giving v = +|f| (1); Image is behind the mirror, so it is virtual (1); m = -v/u = +2 (1); Erect and magnified twice (1).	1+1+1+1+1
	OR: Plane: virtual, same size (1); Concave: real or virtual, depends on object distance (1); Convex: virtual, erect, diminished (1); Field of view: convex is widest (1); Uses: dressing table, headlights, rear-view (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.
- Using f in centimetres while calculating power; $P = 1/f$ gives dioptres only when f is in **metres**.
- Saying that light bends away from the normal when entering glass; it bends towards the normal in a denser medium.
- Describing the image formed by a concave mirror when the object is between P and F as real; it is virtual, erect and enlarged.
- Writing the refractive index as $\sin r / \sin i$; Snell's law gives $n = \sin i / \sin r$ for light going from air into the medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **Light - Reflection and Refraction** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10 cm; less than 10 cm	f = 1/10 m = 10 cm.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Concentration of heat.
3	**(a)** between the optical centre and the focus	Only then is the image virtual, erect and magnified.
4	**(b)** the same focal length as the original	Focal length unchanged; image dimmer.
5	**(d)** A is false but R is true.	A is false; R is true.
6	**(a)** beyond 2F	Object beyond 2F gives a real, inverted, diminished image between F and 2F.
7	**(a)** parallel	Source at the focus gives parallel rays.
8	**(c)** A is true but R is false.	A is true; R is false: frequency remains constant.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$P = 1/f$, so negative f gives negative power.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$P = 1/0.5 = +2$ D; R is a true fact about units and does not explain the value.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Number = 360/90 - 1 = 3 (1); Reflections from both mirrors form 3 images (1).	1+1
12	Apparent depth = real depth / n (1); 3/1.33 = 2.26 m (1).	1+1
13	v = 2×10^8 m/s (1); Frequency remains 5×10^{14} Hz (1).	1+1
14	Virtual images appear only for objects within F (1); In that case the image is always magnified (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Apparent depth = 1.5/(4/3) (1); = 1.125 m (1); The fish appears nearer as rays bend away from normal (1).	1+1+1
16	The whole image still forms (1); Light from the object passes through the lower half (1); Image becomes dimmer because fewer rays pass (1).	1+1+1
17	Convex: u = f, image at infinity (1); Concave: 1/v = -1/15 - 1/15, v = -7.5 cm (1); Concave gives virtual, diminished image; convex gives parallel beam (1).	1+1+1
	OR: Both give real, inverted, same-sized image (1); Mirror image forms on the same side as the object (1); Lens image forms on the opposite side (1).	1+1+1
18	m = +3 = v/u, v = -30 cm (1); 1/f = 1/v - 1/u = -1/30 + 1/10 (1); f = +15 cm (1).	1+1+1
	OR: Concave: magnified erect face (1); Convex: diminished erect face (1); Plane: same size (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $v = c/n = 3 \times 10^8/1.5 = 2 \times 10^8$ m/s (1). (ii) Towards the normal (glass is denser) (1). (iii) $\frac{\sin i}{\sin r}$ = constant for a given pair of media (1). (iv) $\sin r = \frac{\sin i}{n} = \frac{0.75}{1.5} = 0.5$ (1), so $r = 30^\circ$ (1).	1+1+1+2
20	u = 30: v = 60 cm (1); u = 40: v = 40 cm (1); u = 60: v = 30 cm (1); As u increases, v decreases towards f (1); Image size decreases as the object moves away (1).	1+1+1+1+1
	OR: f = -30 cm (1); u = -45: 1/v = -1/30 + 1/45 = -1/90, v = -90 cm (1); m = -2, size 12 cm, real, inverted (1); u = -20: 1/v = -1/30 + 1/20 = 1/60, v = +60 cm (1); m = +3, size 18 cm, virtual, erect (1).	1+1+1+1+1

Mistakes to avoid:

- Ignoring the sign convention: for a concave mirror or lens use $u < 0$ and, for a concave mirror, $f < 0$ (convex lens $f > 0$, concave lens $f < 0$).
- Using the lens formula $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ for mirrors, or the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ for lenses.

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